

Week 12

Building Transparent and Reproducible Data Pipelines

- Certificate IV in Information Technology (Data Science and AI)
- North Metropolitan TAFE, East Perth

Agenda This Week

- Why reproducibility and transparency matter in modern AI
- Key principles of reproducible pipelines
- Industry standards and best practice
- Tools for building and documenting data workflows
- Step-by-step pipeline example in Jupyter
- Activity; documenting your own workflow
- Industry case study
- Assessment preparation and next steps

Learning Objectives

By the end of this week, you will be able to

- Describe principles of transparent and reproducible data workflows
- Construct a simple data processing pipeline using Jupyter Notebooks
- Apply best practice documentation and versioning
- Assess a pipeline for transparency and completeness

Why Reproducible & Transparent Pipelines Matter

- AI models rely on high-quality data and clear workflows to ensure trust
- Poor documentation leads to wasted time, errors, and compliance risks
- Industry requires data processes that can be verified, audited, and repeated

Key Principles for Data Pipelines

- Transparency; Every processing step is visible and understandable to others
- Reproducibility; Anyone can repeat your results with the same inputs
- Documentation; Workflow and choices are clearly explained
- Version Control; Changes are tracked and recoverable

Industry Standards and Expectations

- Follow open dataset documentation standards such as Datasheets for Datasets
- Use versioning tools (e.g. Git) to track scripts and notebooks
- Provide clear, readable reports in Jupyter or similar tools
- Ensure pipelines comply with data ethics and privacy requirements

Essential Tools for Transparent Workflows

- Jupyter Notebooks; Document code and results together for traceability
- Python libraries; pandas, numpy, matplotlib for data prep and visualization
- Version control; Git and GitHub or GitLab for tracking changes
- Workflow templates; Reusable notebook or markdown templates for pipelines

Example; Simple Data Pipeline in Jupyter

```
import pandas as pd

# Load dataset
data = pd.read_csv('open_data.csv')

# Clean missing values
data = data.dropna()

# Extract features
features = data[['feature1', 'feature2']]

# Output summary
print(features.describe())
```

- Each step should be explained and justified
- Use Markdown cells to annotate code in your notebook

Ensuring Workflow Transparency

- Annotate your notebook; explain why each step is needed
- Use visualizations to show changes after each major transformation
- Keep raw, cleaned, and processed data separated and labeled
- Reference data sources and any external scripts or tools used

Activity; Document Your Own Data Pipeline

- Download the provided notebook template from Moodle
- Select a small open dataset (from UCI or another source)
- Plan and document a pipeline including; loading, cleaning, transforming, summarizing
- Submit your notebook with well-written explanations for each step

Industry Case Study; LAION-5B Dataset Workflow

- LAION-5B is a large open dataset powering image-text AI models such as CLIP
- The team shares processing code, quality checks, and documentation online
- Each step from source acquisition to filtering and split is public and reproducible
- This transparency allows others to trust, reuse, and audit the resource

Common Pitfalls and How to Avoid Them

- Unclear documentation; leads to confusion and mistakes; Use markdown and comments
- Missing code or external resources; breaks reproducibility; Use consistent versioning and storage
- Mixing up data versions; Use clear file naming and version control
- Not explaining design choices; Justify each processing step and reference best practices

Preparing for Assessment

- Document your research and pipeline steps in every project
- Use Jupyter Notebooks for EDA and modelling tasks; ensure all steps are clear
- Review assessment criteria from Week 10 and Week 14 project descriptions
- Ask for feedback during class or appointment if unsure about documentation standards

Recap and Next Steps

- Transparent and reproducible data pipelines are a foundation for trustworthy AI
- Mastering these skills strengthens your ability to deliver value in industry roles
- Next week; Applying Machine Learning basics to open datasets
- For more reading; see 'Datasheets for Datasets' and HuggingFace Dataset documentation

Questions and Discussion

- Any questions about pipeline design, documentation, or tools?
- Share your own experience with data workflows
- Let's review your activity plans before submission

